Related Rates

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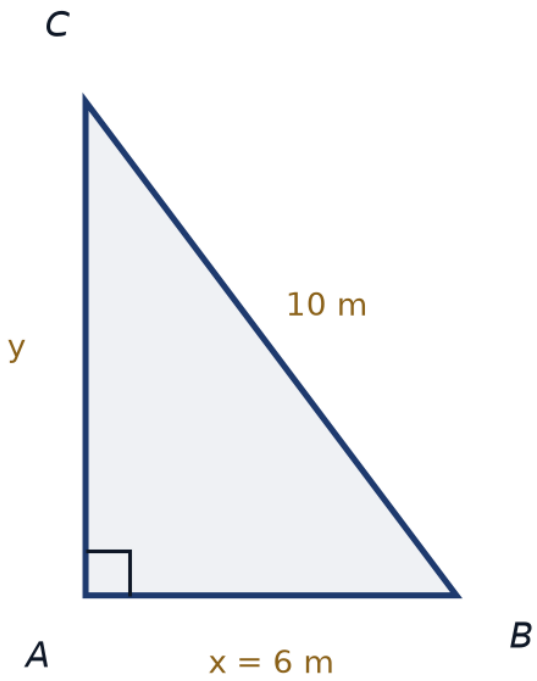
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Setting up related-rates problems**

A circular oil spill is expanding so that its radius increases at 3 m/min. How fast is the area increasing when the radius is 50 m?

2 **[Foundational]** • **Target: Setting up related-rates problems**

A 10 m ladder leans against a wall. The bottom slides away from the wall at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?



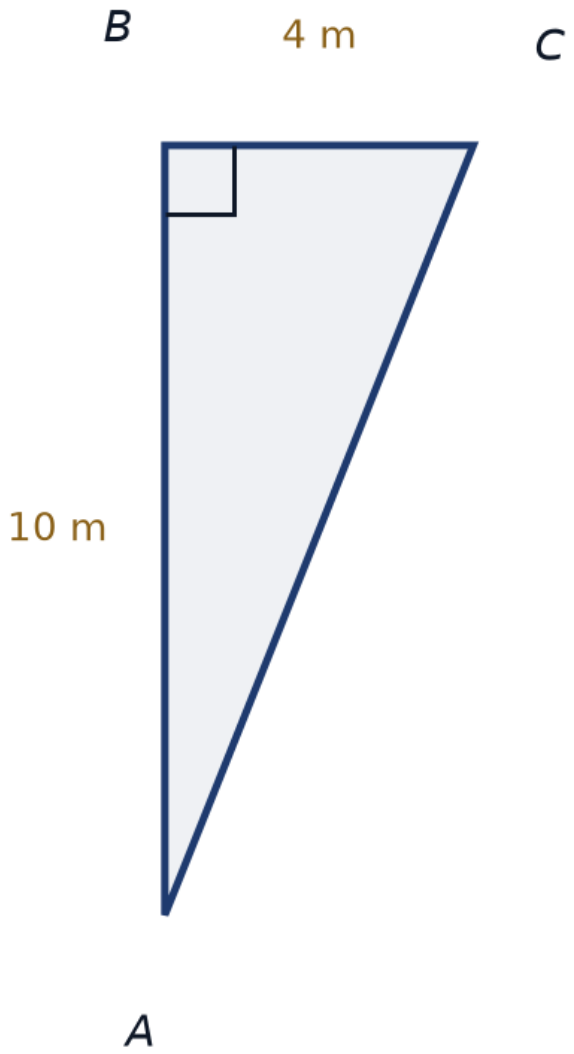
3 **[Foundational]** • **Target: Volume and other rates**

Air is pumped into a spherical balloon at 100 cm^3/s. Find the rate of change of the radius when $r = 5$ cm, to three decimal places.

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4 [Foundational] • Target: Volume and other rates

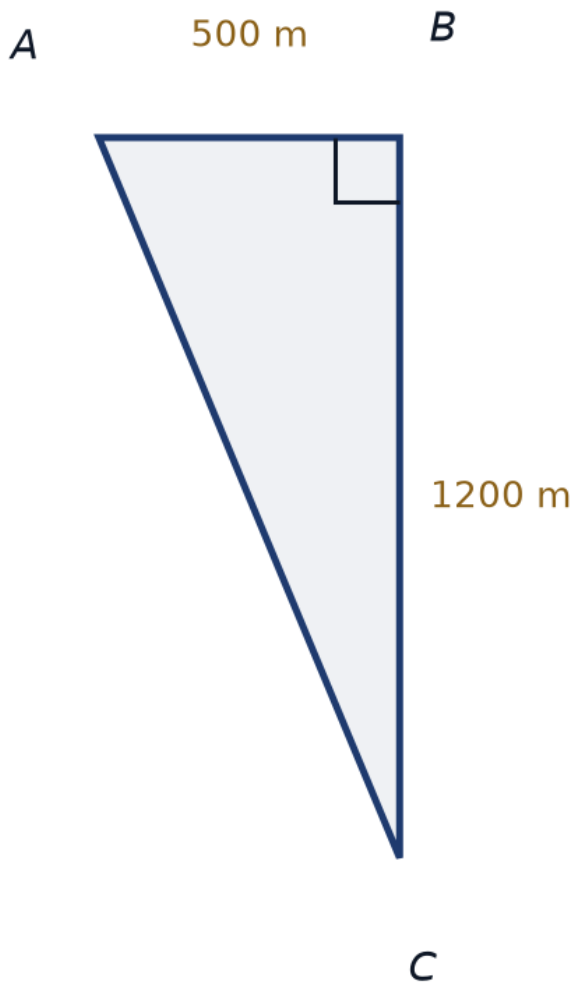
Water drains from a conical tank (vertex down) with radius 4 m and height 10 m at the top, at a rate of 2 m^3/min. Using similar triangles $r = \frac{2h}{5}$, find how fast the water level is dropping when $h = 5$ m.



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5 [Foundational] • Target: Angle and distance rates

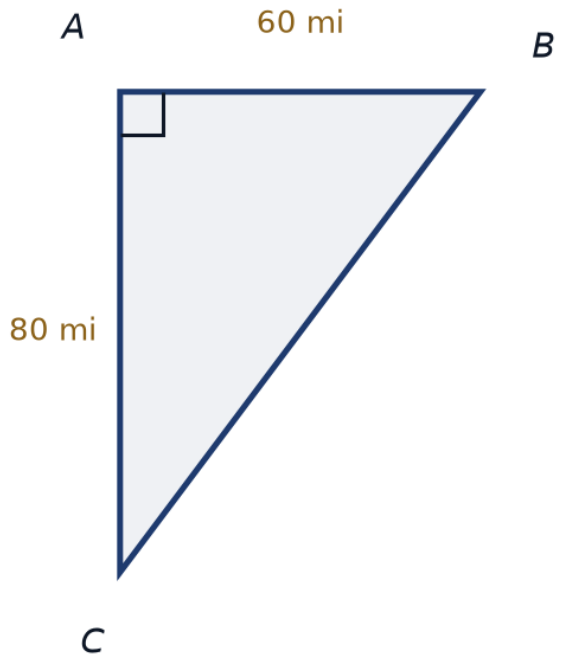
A camera tracks a rocket launching vertically, 500 m from the launch pad. When the rocket is 1200 m high and rising at 200 m/s, how fast is the camera's angle of elevation increasing? Give the answer in radians/s, to four decimal places.



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6 [Foundational] • Target: Angle and distance rates

Two cars leave an intersection at the same time, one heading north at 40 mph and the other heading east at 30 mph. How fast is the distance between them increasing after 2 hours?



7 [Foundational] • Target: More related rates

A streetlight is 6 m high. A 1.8 m tall person walks away from it at 1.5 m/s. Using similar triangles, find how fast the length of the person's shadow is growing.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

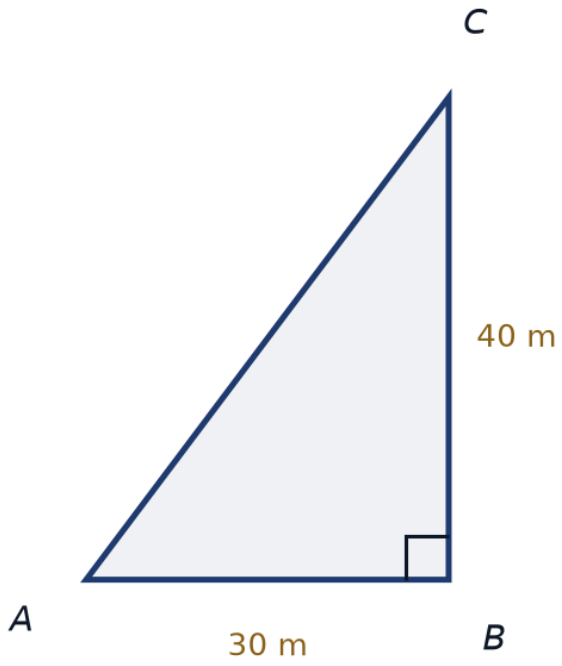
8 [Intermediate] • Target: More related rates

The volume of a cube is increasing at $12\ \text{cm}^3/\text{s}$. Find the rate of change of the side length when the side is $3\ \text{cm}$.

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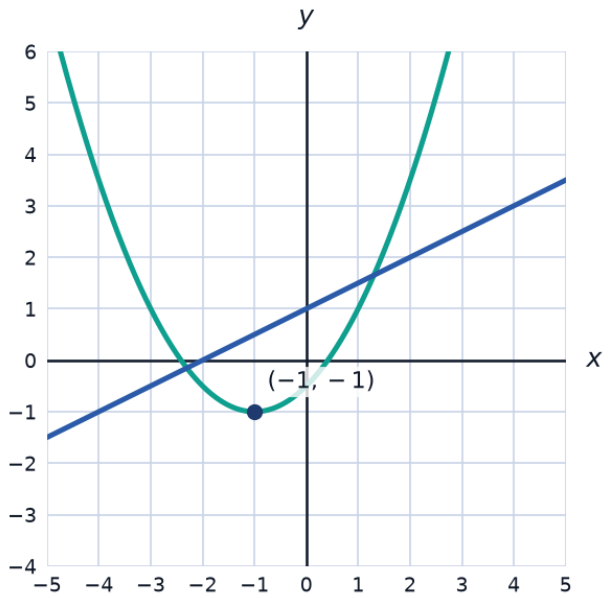
9 [Intermediate] • Target: Rates involving trigonometric relationships

A kite is flying at a constant height of 40 m, moving horizontally away from the person holding the string at 3 m/s. Find the rate at which the string length is increasing when the kite is 30 m horizontally from the person.



10 [Intermediate] • Target: Rates involving trigonometric relationships

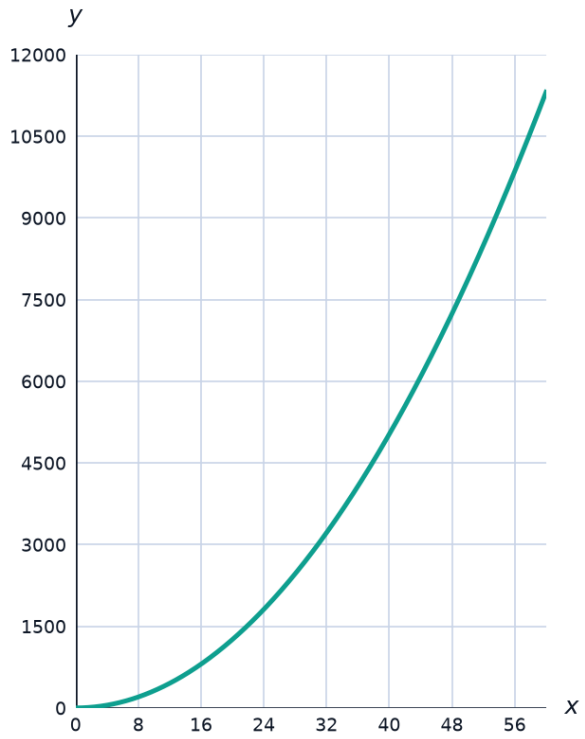
A conical pile of sand has its height always equal to its base radius. Sand is added at $10\ \text{m}^3/\text{min}$. Find the rate of change of the height when $h = 5$ m, using $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$.



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11 **[Intermediate]** • **Target: Visualizing a related-rates scenario**

The graph shows $A(r) = \pi r^2$, the area of the oil spill from the first problem, as a function of radius alone (not time). Explain why $\frac{dA}{dt}$ still depends on $\frac{dr}{dt}$ even though this graph has no time axis.



12 **[Intermediate]** • **Target: Related rates with area and perimeter together**

A rectangle's length is always twice its width. If the width is increasing at 3 cm/s, find the rate of change of the area when the width is 5 cm.

13 **[Intermediate]** • **Target: Related rates with area and perimeter together**

For the same rectangle (length $= 2 \times$ width), find the rate of change of the perimeter when the width is increasing at 3 cm/s.

14 **[Intermediate]** • **Target: A two-stage related-rates problem**

A spherical balloon is being inflated so that its surface area $S = 4\pi r^2$ increases at 12π cm^2/s. Find $\frac{dr}{dt}$ when $r = 3$ cm, then use it to find $\frac{dV}{dt}$ at that instant.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

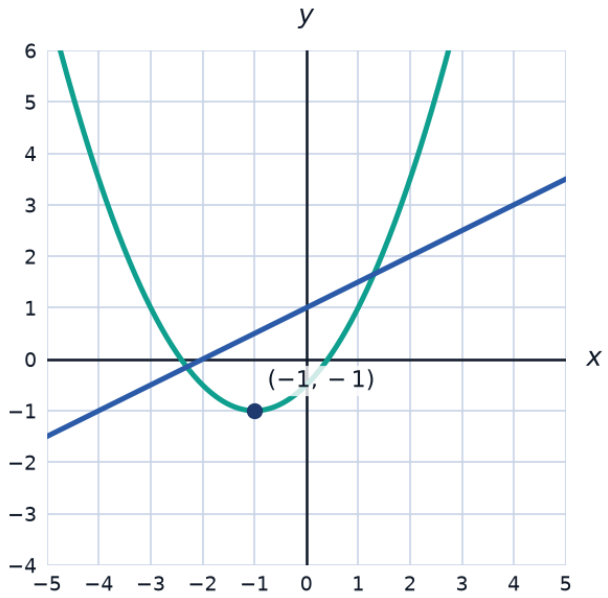
15 **[Challenging]** • **Target: Angle and distance rates** • **Cross-Lesson: 15.01 Double Integrals**

A cone of fixed volume is to be made with minimal lateral surface area. Optimal half-angle α satisfies:.

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16 [Challenging] • Target: More related rates • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Related rates: water drains from a conical tank (height $H = 10$m, radius $R = 5$m) at 2 m³/min. When depth $h = 4$ m, $dh/dt =$.



17 [Challenging] • Target: Rates involving trigonometric relationships • Cross-Lesson: 2.01 Evaluating Limits

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:.

18 [Challenging] • Target: Visualizing a related-rates scenario • Cross-Lesson: 2.02 Continuity

Related rates: sphere volume $V = (4/3)\pi r^3$. If $dr/dt = 2$ cm/s at $r = 3$ cm, $dV/dt =$.

19 [Challenging] • Target: Related rates with area and perimeter together • Cross-Lesson: 3.01 The Derivative from First Principles

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):.

20 [Challenging] • Target: A two-stage related-rates problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Volume of the solid of intersection of two cylinders $x^2 + z^2 \le 1$ and $y^2 + z^2 \le 1$:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Geometric Expansion (Circles, Spheres, Cubes)

The radius of a circle expands at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π cm^2/s

b. 30π cm^2/s

c. 100π cm^2/s

d. 20π cm^2/s

22 [Foundational] • Target: Geometric Expansion (Circles, Spheres, Cubes)

Air is pumped into a spherical balloon so that its volume increases at 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ cm/s

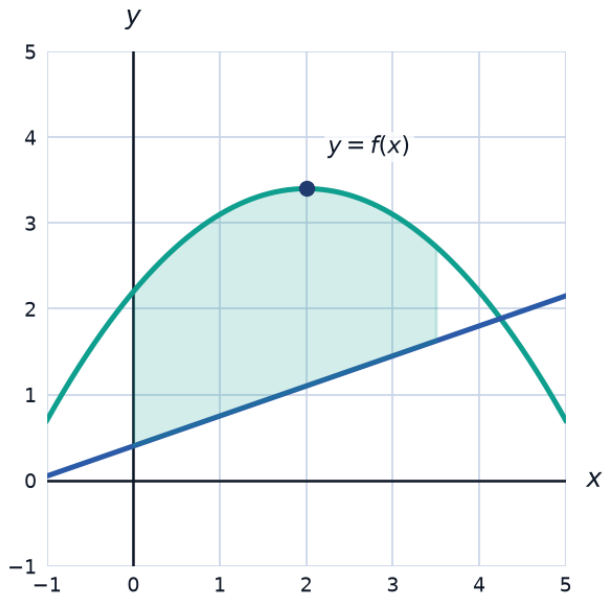
b. $\frac{2}{\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 5π cm/s

23 [Foundational] • Target: Geometric Expansion (Circles, Spheres, Cubes)

The edge length s of a cube increases at 2 cm/s. At what rate is the surface area increasing when $s = 4$ cm?



a. 96 cm^2/s

b. 48 cm^2/s

c. 32 cm^2/s

d. 64 cm^2/s

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24 [Foundational] • Target: Sliding Ladders and Pythagorean Triangles

A $13-$ft ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 ft from the wall?

a. 5/6 ft/s

b. 12/5 ft/s

c. 2 ft/s

d. 1/2 ft/s

25 [Foundational] • Target: Sliding Ladders and Pythagorean Triangles

Two cars start from the same point. Car A travels north at 40 mph and Car B travels east at 30 mph. How fast is the distance between them increasing after 2 hours?

a. 50 mph

b. 70 mph

c. 45 mph

d. 60 mph

26 [Foundational] • Target: Sliding Ladders and Pythagorean Triangles

In the 13-ft ladder problem ($x = 5, y = 12, dx/dt = 2$), at what rate is the area of the triangle formed by the ladder changing?

a. 119/12 ft^2/s

b. 25/2 ft^2/s

c. 0 ft^2/s

d. 12 ft^2/s

27 [Foundational] • Target: Conical and Cylindrical Tanks

Water flows into an inverted conical tank (radius 4 m, height 12 m) at 2 m^3/min. How fast is the water level rising when the depth is 6 m?

a. $\frac{1}{2\pi}$ m/min

b. $\frac{1}{\pi}$ m/min

c. $\frac{2}{\pi}$ m/min

d. $\frac{1}{4\pi}$ m/min

28 [Foundational] • Target: Conical and Cylindrical Tanks

A cylindrical tank of radius 3 m is being filled at 18π m^3/min. How fast is the water height increasing?

a. 2 m/min

b. 3 m/min

c. 6 m/min

d. 1 m/min

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Conical and Cylindrical Tanks**

A trough with triangular cross-section (base 2 m, depth 2 m, length 6 m) leaks water at 0.5 m^3/min. What is dh/dt when $h = 1$ m?

a. $-1/12$ m/min

b. $-1/6$ m/min

c. $-1/8$ m/min

d. $-1/4$ m/min

30 **[Intermediate]** • **Target: Shadows and Angles of Elevation**

A 6-ft tall person walks away from a 15-ft streetlight at 4 ft/s. At what rate is the length of the person's shadow increasing?

a. 8/3 ft/s

b. 2 ft/s

c. 4 ft/s

d. 10/3 ft/s

31 **[Intermediate]** • **Target: Shadows and Angles of Elevation**

In the same streetlight scenario, at what rate is the tip of the shadow moving?

a. 20/3 ft/s

b. 8/3 ft/s

c. 4 ft/s

d. 12 ft/s

32 **[Intermediate]** • **Target: Shadows and Angles of Elevation**

An observer 100 m from a rocket launch site watches a rocket rise vertically at 50 m/s. Find $\frac{d\theta}{dt}$ when the rocket is 100 m high.

a. 0.25 rad/s

b. 0.50 rad/s

c. 1.00 rad/s

d. 0.10 rad/s

33 **[Intermediate]** • **Target: Setting up related-rates problems**

Related rates: a 10-m ladder slides down a wall. When the base is 6 m from the wall and moving away at 2 m/s, the top is moving at:

a. $-\frac{3}{2}$ m/s

b. $-\frac{3}{4}$ m/s

c. $\frac{3}{2}$ m/s

d. $\frac{4}{3}$ m/s

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34 [Intermediate] • Target: Volume and other rates

Optimize area of rectangle with perimeter 20: max area $=$

a. 25

b. 20

c. 50

d. 100

35 [Intermediate] • Target: Angle and distance rates

Related rates: if $A = r^2\pi$, then $\frac{dA}{dt} =$

a. $2\pi r$

b. $2\pi r\frac{dr}{dt}$

c. $\pi r^2\frac{dr}{dt}$

d. $2\pi\frac{dr}{dt}$

36 [Intermediate] • Target: More related rates

Volume of cone $z = \sqrt{x^2 + y^2}$, $z \le 1$ in cylindrical:

a. $2\pi/3$

b. $\pi/3$

c. π

d. $\pi/2$

37 [Intermediate] • Target: Rates involving trigonometric relationships

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$

d. $\pi(\sqrt{2} - 1)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing a related-rates scenario • Cross-Lesson: 2.02 Continuity

A rectangle has perimeter 100. Find the width w that maximises its area.

a. 25

b. $\frac{25}{2}$

c. 50

d. $\frac{100}{3}$

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39 [Challenging] • Target: Related rates with area and perimeter together • Cross-Lesson: 3.01 The Derivative from First Principles

A rectangle has perimeter 120. Find the width w that maximises its area.

a. 40

b. 15

c. 30

d. 60

40 [Challenging] • Target: A two-stage related-rates problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

A rectangle has perimeter 140. Find the width w that maximises its area.

a. $\frac{35}{2}$

b. $\frac{140}{3}$

c. 35

d. 70

41 [Challenging] • Target: Geometric Expansion (Circles, Spheres, Cubes) • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

A rectangle has perimeter 160. Find the width w that maximises its area.

a. 40

b. 80

c. $\frac{160}{3}$

d. 20

42 [Challenging] • Target: Sliding Ladders and Pythagorean Triangles • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

A rectangle has perimeter 132. Find the width w that maximises its area.

a. 66

b. $\frac{33}{2}$

c. 44

d. 33

43 [Challenging] • Target: Conical and Cylindrical Tanks • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

A rectangle has perimeter 148. Find the width w that maximises its area.

a. $\frac{37}{2}$

b. 74

c. $\frac{148}{3}$

d. 37

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44 [Challenging] • Target: Shadows and Angles of Elevation • Cross-Lesson: 6.02 Optimization and Curve Sketching

A rectangle has perimeter 156. Find the width w that maximises its area.

a. 39

b. $\frac{39}{2}$

c. 52

d. 78

45 [Challenging] • Target: Setting up related-rates problems • Cross-Lesson: 7.01 Integration by Substitution

Volume under $z = 1$ over region R of area A is:

a. A^2

b. A

c. $A/2$

d. πA

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